

What Preprocessing Does to Legal Retrieval

Matched-volume controls for recurring legal text and index age

Evidence from published Saudi commercial judgments

Abstract. Legal AI pipelines often remove recurring text and freeze historical indexes, but a downstream score change can confound the semantic intervention with the amount of evidentiary material removed. We test both confounds on published Saudi commercial judgments. The retrieval task is factual rather than synthetic: given the court's reasoning immediately before a statutory citation, retrieve the article the court then cited. Across 105,575 resolved citations and 976 candidate articles, we run the same BM25 retriever under temporally fenced corpus treatments. Removing recurring legal formulas reduces the index by 31.4% and lowers MRR@10 by 0.0241, whereas randomly removing the same number of contexts lowers it by 0.0089. The targeted loss is 2.7 times the mean volume-only loss and, in the five largest-removal folds, is worse than all 20 matched random draws. Yet the same targeted removal is volume-equivalent on a separate corpus-level authority-support analysis, showing that the effect of preprocessing is task-dependent. We apply the same matched-volume logic to index staleness. At one, two, and four quarters, 70%, 64%, and 62% of the observed MRR loss remains after controlling for index shrinkage. The result is a domain-specific evaluation recommendation, not a new ablation method: legal-AI preprocessing should be evaluated against a same-volume control before performance changes are attributed to the class of text removed.

Keywords. legal information retrieval; corpus construction; deduplication; temporal generalisation; evaluation validity; Saudi law

1. Introduction

Between a corpus of published judgments and a legal AI system built on it sit preprocessing decisions that are made once, justified briefly, and then inherited by everything downstream. The most common of them is the removal of recurring text. It is justified by a reasonable-sounding rule: the removed material is boilerplate, and boilerplate is noise.

A downstream score change alone is insufficient to attribute an effect to the semantic class removed. Removing recurring passages also removes a large volume of data, so an intervention evaluated without a quantity control leaves those two effects entangled.

The control is one experiment: remove the same *number* of items at random and see whether the change survives. It is not our idea — matched-budget random subsets are established in the data-selection literature (§2) — and we make no claim to it. We test what that control reveals when the removed material is legal evidence rather than generic web text, and use the result to motivate a domain-specific evaluation recommendation.

The question. Does targeted removal of recurring legal wording change legal retrieval performance beyond what removing the same amount of corpus material does anyway? And separately: how much of what a stale index costs is age, rather than the stale index simply being smaller?

The instrument. A legal retrieval task built from what courts actually did: given the court's own reasoning in the run-up to a statutory citation, retrieve the article it then cited (§5). 105,575 resolved citations, 976 candidate articles, one relevant article per query, temporally fenced, with every citation span inside a query masked. The task is deliberately standard and the retriever is deliberately simple — BM25 — so that the corpus treatment remains the principal experimental contrast.

What we find.

- **Targeted recurrence removal is not volume-equivalent on the retrieval task.** Removing every circulating formula costs 2.7 times what removing the same number of contexts at random costs, and lands outside the spread of 20 size-matched random draws in the five largest-removal folds (§6). The removed contexts are disproportionately useful to this retrieval instrument.
- **On a different downstream task in the same corpus, the same removal is volume-equivalent** (§6.4). Two tasks therefore give different answers about whether the targeted removal is special; the matched-volume arm distinguishes those cases in this design.
- **Roughly two thirds of the measured stale-index loss remains after a same-volume control** (§7). Temporal decay and controlled decomposition are prior art; the contribution here is the specific age-versus-shrinkage estimand on this corpus.
- **Adding the parties' own citations to the index grows it by 11 per cent, raises the recall ceiling, and lowers MRR** (§7.3).

What we contribute, stated so a reviewer can check the boundary:

1. A controlled legal-retrieval result showing that removal of recurring legal wording costs **more** than removal of the same evidentiary volume (§6).
2. A decomposition of stale-index loss into the part explained by a smaller frozen index and the residual age effect (§7).
3. A measurement of what the removed material actually is — its unit, its volume, and what it does to authority support (§4) — which is what makes the intervention worth testing at all.
4. A domain-specific validity recommendation follows from those experiments, but the control itself, downstream evaluation of cleaning, the retrieval task, and temporal drift are all prior art (§2).

Everything is computed in a version-controlled research repository with a determinism check, a figure-tracing guard and frozen baselines; §11 gives the reproduction path.

2. Related work and positioning

Five lines of work meet here.

Deduplication of training corpora. Removing exact and near duplicates is standard practice, motivated by memorisation and by the distortion of frequency statistics (Lee et al. 2022). The closest methodological warning is Schäfer et al. (2024): merging naturally occurring near-duplicate subwords **hurt** language modelling, and the authors calibrated that loss against baselines trained on 95 and 90 per cent of the data, describing the cost as roughly equivalent to 5–10 per cent less training

data. That makes the quantity-confound principle prior art. Our question is narrower: whether a legal-corpus intervention that removes evidentiary contexts has a downstream retrieval effect beyond a *random removal of the same contexts count*.

Boilerplate removal as a retrieval intervention. Web2Text (Vogels et al. 2018) and later perplexity-based cleaning (Fernández-Pichel et al. 2024) already evaluate boilerplate removal through downstream information retrieval. Web2Text is especially relevant: low-recall cleaners reduce retrieval quality because relevant text is removed, whereas more accurate cleaning can improve it. Thus neither "cleaning affects retrieval" nor downstream IR validation is new here. What those experiments do not provide is a matched-volume random removal that asks whether the observed change is special to the removed class.

Data selection with matched-budget controls. Comparing a curated subset against a randomly drawn subset **of the same size** is established methodology in data-selection work (Li et al. 2024). The control in §6 is imported, not invented. The legal-domain recommendation in §6.5 is therefore a synthesis supported by the experiment, not a standalone methodological-first claim.

Legal citation-context retrieval. Retrieving a cited authority from the text around the citation is an established task (Huang et al. 2021), with temporally fenced variants that measure the leakage controlled here (Liu et al. 2026). For this jurisdiction, Abu Shairah et al. (2025) already define regulation identification over 13K Saudi commercial cases in ALARB. Our task is therefore an **instrument, not a contribution**: it is deliberately standard because the retrieval architecture is not the object of study.

Temporal drift and index maintenance. Persistence of retrieval systems over a moving corpus has venue-scale evaluation infrastructure. LongEval 2024 compares NDCG/MAP over evolving collections; Ovcharov (2026, arXiv:2605.17639) measures two decades of legal statute-retrieval decay and uses fixed-article and temporal train/test ablations to separate decay from composition/evaluation artefacts; and Liu, Tan and Liu (2026, arXiv:2607.17142) decompose a legal citation-context retrieval gain into future-evidence leakage, legitimate admission cost and an index effect. Thus neither ageing nor controlled temporal decomposition is new: the closest legal neighbours already separate temporal leakage, admission effects, index effects, and longitudinal decay under explicit time controls. Section 7 reports a different estimand for this corpus: when a BM25 index is frozen at an earlier quarter, how much of its MRR loss is reproduced by randomly removing the same number of contexts from the current index, and how much remains associated with age.

Our position is that these five lines constrain the claim rather than decorate it. The paper does **not** introduce deduplication, downstream evaluation of boilerplate removal, matched-budget controls, citation-context retrieval, or temporal persistence. It places those established ideas into one legal-corpus experiment in which the evidentiary material removed and the time at which it was available are themselves part of what the retrieval system is supposed to represent.

Note on references. Citations above are given by name and year rather than as a formatted list.

REFERENCES_TODO.md records the reading state and claim consequence of each source. The two references load-bearing for the quantity and temporal controls (Schäfer et al.; LongEval 2024) have now been read in full, and the relevant retrieval sections of the closest boilerplate work have been checked. Sources still marked abstract/snippet may not carry a novelty claim in the submitted version.

3. Data and setting

The corpus. Published first-instance and appellate commercial judgments of the Saudi Ministry of Justice. The AI-subject scan covers **50666** judgments; the measurement layers used here cover the hijri years 1442–1446, from which we extract **25213** non-statutory authority mentions across **27027** judgments carrying a reasoning section, and **105575** resolved statutory citations with a usable preceding context — **47492** of them in the court's own reasoning, over **976** distinct (instrument, article) targets across **68** instruments. The last three figures are the retrieval task of §5.

This corpus family is not unexplored. Alharbi et al. (2026) publish ArabiCCR, a dataset of Saudi commercial rulings from the same Ministry gateway, and Abu Shairah et al. (2025) define ALARB over 13K Saudi commercial cases (§2). We do not claim a new dataset; we claim controlled measurements on one.

Why this setting. Saudi commercial judgments carry an unusually wide range of authority in one document — statute, the parties' contract, named works of fiqh, maxims of fiqh, Qur'an, hadith, settled judicial principle, commercial custom, and the court's own named discretion — and the published documents carry structural headings that separate the recital of the parties' claims from the court's reasons. Both properties are instrumentally useful: the first gives a rich authority space in which redundancy is visible; the second lets every measurement be computed in the court's own voice rather than over a blend of voices.

Layers. Five derived layers are used, and none of them stores judgment text: a statutory citation layer (instrument and article, by speaker), an authority mention layer (nine authority types, by speaker), a canonical identity layer for non-statutory authority (28 canonical identities plus generic classes), the formula layer defined in §4, and the retrieval layer defined in §5. The first four record counts and identifiers only. The fifth records the material BM25 consumes — an unordered bag of hashed tokens per context, word order destroyed and no token recoverable — which is the weakest representation that supports the experiment and still cannot be read back as a passage.

Time index. Hijri quarters, 1442Q1–1446Q2, of which ten satisfy a maturity rule fixed before any outcome was inspected: at least 800 judgments carrying court authority, at least 200 court statutory citations, a later quarter must exist, and volume at least 40 per cent of the median of the preceding four quarters. All time-indexed results below are computed on mature quarters only.

What the corpus is not. It is published commercial adjudication. Absence from it is absence from that record, never absence from Saudi law. Two of the most active current legal frontiers in the jurisdiction — AI governance and data protection — are regulated in forums this corpus does not contain, a point we return to in §9.

4. What recurs, and what it is worth before anything is built

The intervention tested in §6 removes recurring wording. This section says what that wording is, how much of the corpus it is, and what removing it does to a frequency count — the corpus-level facts that make the intervention worth testing, and that a practitioner would use to justify it.

4.1 The unit

We define an **authority-adjacent recurring formula** as the first 12 hex characters of the SHA-1 of a normalised window of **90** characters on each side of a matched authority mention. Normalisation removes diacritics, folds orthographic variants, collapses whitespace, deletes every character outside the Arabic letter class — so digits and punctuation are removed — and deletes words of one or two characters. The surviving token string has a median length of **29** tokens (p10 26, p90 33).

Three properties of this unit matter and are stated because they bound every count below. It is **exact**: a cryptographic hash has no neighbourhood, so a single differing surviving word is a different formula. It is **source-preserving**: the matched authority string lies inside the window. And it is **not a representation of a judgment's language**: it is a fixed-width neighbourhood of an authority mention, and no claim in this paper is about judicial writing in general.

We also built a near-duplicate layer — a banded minhash over token 3-shingles (Broder 1997), 8 bands of 4 — and set it aside: of 130 pairs grouped at Jaccard 0.7, only **0.5462** survive at 0.8. The grouping does not survive a change of threshold at this corpus size, so every count below is exact-match and is therefore a floor.

4.2 How much redundancy there is

Across 25213 mentions there are **14958** distinct formulas. **218** of them recur in ten or more judgments, accounting for **5981** mentions.

Masking the matched authority string before hashing reduces the distinct count only from 14958 to **14941**, and **0** of the 218 circulating formulas carry more than one canonical authority identity. At this resolution, recurrence is source-bound: there is no observed shell that recurs ten times and receives different authorities. We record this as an absence of observation rather than a demonstrated impossibility, since the near-family layer that would catch a shell varying by one word is the layer we could not stabilise.

4.3 What the redundancy does to support

Define **raw support** for an authority as its mention count, and **formula-adjusted support** as the count in which each circulating formula contributes once per authority and every non-recurring mention contributes once. Adjusted support is a floor on the number of independent contexts, not an estimate of it.

	raw	adjusted	inflation
all 34 authorities	25213	19450	1.296
Ibn Taymiyya	3129	1660	1.885
al-Insaf	884	413	2.14
unattributed fiqh	4783	3601	1.328
untraced hadith	5009	4088	1.225
worst single authority (al-Bayhaqi)	—	—	2.429

A frequency-trained or frequency-ranked system over-weights the most formulaically cited authorities by up to a factor of **2.429**, and by **1.296** on average.

This is invisible to document-level deduplication, which is the standard remedy. The formulas recur *across* documents, not within them: each judgment contains the passage once, and the judgments are not duplicates of one another. No document-level or near-duplicate-document filter removes any of it.

4.4 What it does to ranking, and what it does not do to trend

Ranking authorities by adjusted rather than raw support moves **21** of 34 at least one place; the largest displacement is 5 places (al-Bayhaqi). Top-10 stability is **0.9**: one authority enters the top ten under deduplication (muttafaq 'alayh) and one leaves it (al-Insaf).

The temporal shape, by contrast, is essentially untouched: the raw and adjusted quarterly series correlate at **0.9976**, and the inflation ratio moves only from **1.0846** in the first mature quarter to **1.099** in the last.

The engineering reading is specific. **Formula redundancy changes the level and the ranking of apparent support and not its shape over time**. A system that ranks authorities is affected; a system that reports a trend is not. A practitioner who needs only the trend can ignore this; one who ranks, weights or retrieves cannot.

4.5 What the recurring formulas are

Each circulating formula is assigned a class from a coarse mechanical taxonomy built from keyword markers with a fixed priority — no model, no learned labels, and no reading of any passage. The classes are dominated not by procedural boilerplate but by the framing of authority:

class	formulas
authority introduction frame	46
generic reasoning	46
authority quotation	43
compensation and harm	36
burden and presumption	21
doctrinal rule	9
disposition	5
procedural operation	5
contract	4
jurisdiction	2
fact recital	1

Procedural boilerplate is 5 of 218. Anyone deleting this material as boilerplate is mostly deleting the sentences in which courts introduce, quote and apply authority. §6 measures what that costs a retriever.

5. The experiment

5.1 The task

Given the court's own reasoning in the 600 characters before a statutory citation, retrieve the article the court then cited.

Nothing in it is synthetic. The query is text a Saudi commercial court wrote. The label is the citation that court then made, resolved by the same instrument matcher every other measurement in this paper uses. Relevance is never judged by us, and there is exactly one relevant article per query.

A corpus pass over the five hijri years yields **105575** resolved statutory citations carrying a usable context window, of which **47492** are in the court's own reasoning, over **976** distinct (instrument, article) targets across **68** instruments. No judgment text is stored: the context is written out as an unordered bag of hashed tokens, which is all BM25 consumes and which cannot be read back as a passage.

The task is **not a contribution**. Retrieving a cited authority from its own local context is established (§2), and a regulation-identification task already exists for this jurisdiction. It is used here as an instrument so that controlled corpus treatments can be compared on the same task.

5.2 The retriever

BM25 (Robertson and Zaragoza 2009), $k1 = 1.2$, $b = 0.75$, over one pseudo-document per article, pooled from the contexts in which earlier judgments cited that article. The retrieval architecture and query sample are held fixed across the corpus-treatment arms. A dense retriever would answer the additional question of architecture dependence and is not run; nothing here is a claim about what a neural retriever would do.

One efficiency device exists — postings whose idf falls below a floor are dropped — and it is reported because it **never fired**: the dropped share is 0.0 in every fold, so no approximation is in force in any number below.

5.3 Leakage

Five controls are structural, applied when the layer is extracted and before any split exists, and one is an experiment in its own right.

1. The query window ends where the citation begins, so the answer is never inside its own query.
2. Every statutory citation span **inside** a query window is masked. A second citation of the same article nearby cannot leak it, and the instrument's name cannot either.
3. The window is clipped to its segment: a citation in the reasons never reads the recital.
4. The split is temporal. The index holds strictly earlier quarters, so a query's own judgment can never be in the index.
5. Circulating formulas are recomputed **on each fold's own index**, never on the whole corpus — reading recurrence off the full corpus would let a fold see its own future.
6. **RAW_NO_FP_LEAK** prices what remains: drop every index context whose fingerprint also occurs in a query this fold.

Control 6 is not a formality. It costs **0.0643** MRR@10 — more than de-boilerplating costs. Verbatim recurrence between index and query is a real part of what this retriever scores, and every figure below should be read as including it except the RAW_NO_FP_LEAK row itself.

5.4 Folds and metrics

Ten folds, on `horizon.py`'s SCORABLE quarters — the maturity rule fixed before any outcome here was inspected and reused unchanged. Queries are capped at **1000** per fold, sampled once with seed **20260901**, and **shared by every arm**, so arms are compared on identical queries.

Recall@1, @5, @10 and MRR@10, micro-averaged within a fold and then averaged over folds, reported beside the index size and the share of queries whose gold article is in the index at all — the ceiling on recall for that arm. nDCG is omitted deliberately: with one relevant document per query it is a monotone transform of the reciprocal rank.

5.5 What is varied

Only the index.

arm	index
RAW	every court-reasoning context from earlier quarters
FORMULA_DEDUP	minus contexts sitting in a circulating formula
MATCHED_RANDOM	minus the same <i>number</i> of contexts, at random, 20 seeded draws
FROZEN_kQ	cut back k quarters — corpus ageing
FROZEN_kQ_VOLUME	minus the number of contexts freezing removed, at random, 10 seeded draws
PLUS_PARTY	court contexts plus the parties' own citation contexts
RAW_NO_FP_LEAK	RAW minus any context whose fingerprint occurs in a query

5.6 Every arm, every metric

Means over folds, so that nothing quoted later has to be taken on trust. `gold in index` is the share of queries whose gold article is in the index at all — the ceiling on recall for that arm.

arm	R@1	R@5	R@10	MRR@10	gold in index	contexts	articles	folds
RAW	0.5026	0.7556	0.8105	0.6136	0.9248	18673	504	10
FORMULA_DEDUP	0.4708	0.7462	0.8093	0.5895	0.9229	12805	498	10
MATCHED_RANDOM	0.4946	0.7492	0.8027	0.6047	0.9188	12805	446	10
PLUS_PARTY	0.4994	0.7542	0.8115	0.6102	0.9318	20712	612	10
RAW_NO_FP_LEAK	0.4278	0.7115	0.7789	0.5493	0.9089	15992	501	10
FROZEN_1Q	0.4422	0.6921	0.7474	0.5491	0.8915	15059	432	10
FROZEN_1Q_VOLUME	0.4857	0.7374	0.7923	0.5942	0.9086	15059	450	10
FROZEN_2Q	0.3872	0.6224	0.6759	0.4875	0.821	11964	361	10
FROZEN_2Q_VOLUME	0.4619	0.7085	0.7654	0.5685	0.882	11964	391	10
FROZEN_4Q	0.3202	0.5394	0.5929	0.4146	0.7232	7685	267	9
FROZEN_4Q_VOLUME	0.4282	0.6841	0.7431	0.5373	0.8582	7685	307	9

FROZEN_4Q and its control score 9 folds rather than 10: the earliest fold has no index left four quarters back.

6. Result 1: de-boilerplating costs more than its volume

6.1 The headline

arm	R@1	R@5	R@10	MRR@10	gold in index	contexts	articles
RAW	0.5026	0.7556	0.8105	0.6136	0.9248	18673	504
FORMULA_DEDUP	0.4708	0.7462	0.8093	0.5895	0.9229	12805	498
MATCHED_RANDOM	0.4946	0.7492	0.8027	0.6047	0.9188	12805	446

Removing the circulating formulas takes the mean index from 18673 contexts to 12805 — **31.4** per cent of it — and costs **0.0241** MRR@10. Removing the same number of contexts at random costs **0.0089**. The targeted removal is **2.7 times** the price of the volume it removes.

6.2 The per-fold test

A difference of two means is the weak version of this. Each fold has its own distribution of 20 size-matched random draws, and the question is whether the targeted removal lands outside it.

fold	circulating formulas	contexts removed	share	outside the random spread	draws beaten, of 20
1443Q1	0	0	0.0	removed nothing	—
1443Q3	7	101	0.1132	no	6
1443Q4	33	690	0.2286	no	1
1444Q1	64	1402	0.2675	no	13
1444Q2	133	3203	0.2808	yes	0
1444Q3	228	5921	0.2959	no	1
1444Q4	307	8178	0.3099	yes	0
1445Q1	372	10416	0.321	yes	0
1445Q4	479	13973	0.3297	yes	0
1446Q1	500	14792	0.3313	yes	0

DEDUP_EFFECT_EXCEEDS_VOLUME_EFFECT. The targeted removal sits inside the random spread in 4 of the 9 folds in which it removed anything, and those are the four smallest removals; in the last five folds it beats **0 of 20** draws every time. Across live folds it beats **2.3333** draws on average out of 20.

There is a dose-response shape here that we did not design for: as the index grows and the removal share rises from 0.113 to 0.331, the targeted removal moves from indistinguishable from random to worse than every draw.

6.3 What that means about the removed text

The recurring wording is disproportionately useful retrieval evidence on this task. A recurrence-based cleaning rule therefore removes material whose retrieval value is higher than a same-volume random sample from the index.

The shape of the disagreement is worth one sentence more, because it is the opposite of what a practitioner would guess. Deduplication keeps **498** of 504 articles while random removal keeps **446** — the targeted removal *preserves the article inventory better* and still loses more. What it takes is not coverage. It is the evidence per article.

This is consistent with what §4 measured: the circulating formulas are dominated by authority-introduction frames and quotations rather than by procedural boilerplate, with procedural operation only 5 of 218. Removing them removes the sentences in which courts introduce and apply authority, which is exactly the material a retriever built on citation contexts needs.

6.4 The same removal, a different task, the opposite answer

The same corpus supports a second downstream question with a binary verdict: among non-statutory authorities first observed beside a given code, do those first observed in the court's voice persist better than those first observed in the parties'? On the full data the matched comparison returns a court-first advantage; removing the **5981** mentions in circulating formulas flips it.

We ran the same two controls there. Eleven class-specific removals — one class of formula at a time — reproduce the flip **not once**. And a size-matched random removal reproduces it at every level tested, 20 seeded draws each:

removed	mean mentions	removed flip share
25 per cent	1527.2	0.1
50 per cent	3017.1	0.3
75 per cent	4514.0	0.7
90 per cent	5407.1	0.9

On that task the targeted removal is volume-equivalent: the matched comparison rests on 6 or 7 pairs in every arm, and a comparison that thin moves when a quarter of the evidence leaves, whatever leaves.

So: one corpus, one intervention, two downstream tasks, opposite answers about whether the intervention is special.

The pair shows that whether removal is volume-equivalent is a property of the *intervention and the task together*, not of the intervention alone. In this design, the matched-volume arm is what separates a class-specific retrieval effect from a quantity effect.

6.5 The evaluation recommendation

Matched-budget and quantity-calibrated controls are established methodology in data selection and deduplication (§2); we are importing that logic, not inventing it. The legal-retrieval result motivates the following domain-specific recommendation:

When a preprocessing intervention removes substantial evidentiary volume from a legal corpus, its effect on legal-AI evaluation should be compared with a matched-volume random-removal control before the change is attributed to the semantic class of text removed.

The cost is one extra arm and a seed. In this study, that arm changes the interpretation of the same preprocessing intervention across two downstream tasks; without it, the semantic and volume effects would remain conflated.

7. Result 2: what index age costs, once shrinkage is taken out

7.1 The confound

Freezing an index makes it two things at once: older and smaller. Prior work already decomposes temporal retrieval effects in several ways (§2), but the specific shrinkage confound remains in a simple frozen-index comparison. We therefore pair each frozen index with a live index from which the same number of contexts is removed at random, and report the residual loss associated with age.

index frozen by MRR@10 loss vs RAW of which volume of which age age share

1 quarter	0.0645	0.0194	0.0451	70 %
2 quarters	0.1261	0.0451	0.081	64 %
4 quarters	0.199	0.0763	0.1227	62 %

Under this matched-volume comparison, roughly two thirds of the measured loss remains associated with age rather than shrinkage. Section 8 keeps the publication-composition confound explicit, so this residual is not interpreted as a pure causal effect of elapsed time.

The recall ceiling moves with it. gold in index falls 0.9248 → **0.8915** → **0.821** → **0.7232** as the index is frozen by one, two and four quarters, while the matched-volume controls hold at **0.9086**, **0.882** and **0.8582** — so most of the vanishing coverage is articles the court had not yet begun to cite, not articles thinned out by a smaller index.

7.2 Ranking ages faster than recall

The same ageing seen from the ranking side, on the retrieval snapshots the foresight layer freezes rather than on this BM25 index:

horizon	citation share to never-seen articles	top-50 displacement (%)	mean rank displacement, top 200
1 quarter	0.0437	34.9231	33.1885
2 quarters	0.0612	39.1667	36.3225
4 quarters	0.1191	46.4	40.513

At one quarter the index is still covering the law — only 4.37 per cent of citation mass goes to articles it has never seen — while more than a third of its top fifty is no longer in the court's top fifty. Against thresholds fixed in advance (30 per cent displacement, 35 rank places, 10 per cent content gap), the displacement trigger crosses at one quarter, the rank-gap trigger at two, and the content trigger at four. **A maintenance policy calibrated on coverage refreshes far too late.**

7.3 Staleness is a clock, and party-side material does not pay

Two negative results that a builder can act on.

Refresh triggers. Applying the same staleness criterion at pseudo-event dates on instruments already mature when the window opened, the criterion fires in **15 of 15** pseudo-events with no legal event at all, and no staged multi-layer pattern appears in them (share **0.0**). Staleness is a property of elapsed time. Event-triggered refresh cannot be justified against periodic refresh on this evidence, and we record the proposal as held rather than adopted.

High recall. Adding the parties' own citation contexts to the index grows it by **11** per cent (18673 → 20712 contexts, 504 → 612 articles), raises the recall ceiling from 0.9248 to **0.9318**, and **lowers** MRR@10 from 0.6136 to **0.6102**. Measured a second way over 13 folds, the party-only remainder grows the universe by **40.6** per cent, adds **0.0064** of coverage, and **0.0956** of what it adds is ever cited by a court — **0.157** coverage points per 10 per cent of universe growth. **HIGH_RECALL_COSTS_MORE_THAN_IT_BUY.**

Which index composition drifts least. Compared on drift rather than point-in-time coverage over 5 pseudo-future folds:

architecture	mean coverage	drift
statute + doctrinal companions	0.953	-0.0108
statute + current article ecology	0.9417	-0.0169
time-aware recent window	0.9287	-0.0225
statute only	0.9273	-0.0308
speaker-aware hybrid	0.9284	-0.0343

Adding the non-statutory companion layer to a statutory index both raises coverage and roughly thirds the drift of a statute-only index; the index enlarged with party-side material drifts most.

8. Why the temporal fence in §5 is not enough

The experiment splits on time, which is the accepted remedy for temporal leakage. The remedy assumes the two sides of the cut are draws from one process. On this corpus they are not, and the part that moves is not in the text.

Twenty-two series in five independent metric families — publication, docket composition, statutory salience, authority ecology and the formula layer — were scanned with four transparent change-point methods, every statistic scored against a permutation null built from the series' own values, 2000 permutations, fixed seed, $\alpha = 0.05$. The whole battery was then pointed at **200** shuffled redraws of itself to measure what it reports when there is nothing to report.

	observed	null (200 shuffled redraws)
metrics with a significant break	12 of 22	0.11 per metric; mean 2.42 metrics per draw
multi-layer candidate quarters	6	mean 0.365 ; maximum 3 in any draw

Six multi-layer candidate quarters lies outside the entire null distribution. **OBSERVED_EXCEEDS EVERY NULL_DRAW.** A chronological split on this corpus is not a split of one process.

And the moving part is the one a text-only pipeline cannot see. Asked whether any of the six candidates survives removing the two families that describe *what the publisher released* rather than *what courts did*: **none of six**.

family	metrics tested	firing	share
docket composition	6	5	0.8333
formula layer	4	3	0.75
publication	4	2	0.5
statutory salience	4	2	0.5
authority ecology	4	0	0.0

The authority ecology — hybrid rate, named-source share, source concentration, traceability — carries no significant break at all. The standard hope that a target unstable across a window is stable within regimes does not survive testing either: with rolling origins where the break is re-detected on history alone at each origin, segmentation improves forecasting on **0 of 22** series against a last-value baseline.

For this paper the consequence is narrow and should be read narrowly. Our temporal fence prevents a query's own judgment from being indexed; it does not make the folds exchangeable, and the ageing decomposition in §7 is measured across a publication-composition shift that we can detect and cannot remove. Every fold-level figure carries that.

9. The setting: AI is permitted in this procedure and absent from these disputes

This section is context, and we mark it as context. **It does not carry the paper's claim to a venue** — §5 to §7 do that, and would stand if this section were deleted. What it establishes is that the systems the preceding sections are about are not hypothetical in this jurisdiction. Both facts are quoted from enacted instruments rather than asserted.

The forum already permits AI in its own procedure. The implementing regulation of the very statute governing the courts whose judgments we measure provides:

«يجوز الاستفادة من تقنيات الذكاء الاصطناعي في الإجراءات الإلكترونية، ويستغنى عن أي إجراء تحققت غايته باستخدام تلك التقنية» — اللائحة التنفيذية لنظام المحاكم التجارية، المادة الرابعة والعشرون

and the procedural manuals of the Law of Evidence provide:

«يجوز الاستعانة بالتقنيات الحديثة في إجراءات الإثبات، بما في ذلك الذكاء الاصطناعي، ويُستغنى عن أي إجراء تحققت غايته باستخدام هذه التقنيات.» — الأدلة الإجرائية لنظام الإثبات، المادة الثالثة والعشرون

Two further instruments permit AI in notarisation and in the delivery of enforcement services. These are permissions to use AI *in the procedure that produces the record we measure*.

And AI is the subject of none of the disputes. An AI-subject radar over all **50666** judgments returns **0** at the materiality level — no judgment in which an algorithmic or AI feature is shown to be at issue in the dispute — against **28** judgments in which an AI-relevant technology appears without being shown to be at issue, and **12** carrying an explicit AI term anywhere in the document.

The conjunction is the point. **Procedural AI use is legally contemplated; deployment consequences are untested.** These provisions do not authorize any particular retrieval architecture. They establish only that AI-assisted procedure is not a purely hypothetical policy context for the measurements in §5 to §7.

We make no claim about whether any deployed system is affected. Seven verified AI adoption events in this jurisdiction were classified on a linkability ladder and **none** reaches the adjudicatory workflow this corpus observes, so no deployment can be associated with any observable change here. That is a statement about linkability and not about effect, and it is precisely why the corpus-level properties are the part that can be measured today.

10. Limitations

One retriever. BM25 only. A dense retriever would answer a different question — how a learned representation responds to the same corpus treatments — and it is not run. Nothing here is a claim about what a neural system would do.

One jurisdiction, one forum. Published Ministry of Justice commercial adjudication. The dose-response shape in §6.2 is one corpus's, and the direction of the §6 result may not hold elsewhere. The transferable claim is therefore limited to the evaluation recommendation in §6.5, not the observed effect size or direction.

1000 queries per fold. A compute budget, not a design choice. The sample is seeded and shared across arms so it cannot favour one, but a full-query run would narrow the fold-level intervals.

The candidate universe is the extractor's. 976 (instrument, article) targets, bounded by what the instrument matcher resolves; 28 canonical identities bound §4.

Exact-fingerprint resolution. The redundancy unit is exact; the near-duplicate layer built to soften it is unstable at this corpus size (0.5462 pair survival across thresholds). Every redundancy count is a floor, so §6 removes *less* than a fuzzy de-boilerplating step would, and the price it measures is a lower bound on that step's price.

The §6.4 comparison is thin. The matched doctrinal verdict rests on 6 or 7 pairs in every arm. Its thinness is part of the finding — it is why a quarter of the evidence leaving moves it — but it means the volume-equivalent side of the pair is a demonstration that the control is necessary, not a measurement of how often deduplication misleads.

No publication date. The corpus carries a decision date and a retrieval timestamp, and neither institution publishes a per-judgment publication date. Decision-to-publication lag cannot be separated from legal change, which is the confound §8 identifies and cannot remove. It sits underneath §7's ageing figures.

Observation lag. The latest mature quarter is roughly two years behind the time of writing. Every "current" statement is current-as-published.

Nothing is causal. An arm scoring lower is an arm scoring lower. Co-occurrence of wording is not copying.

The taxonomy is mechanical. Formula classes come from keyword presence in a 180-character neighbourhood. No class is a reading of a passage.

11. Reproducibility and AI-assisted research disclosure

All results are computed by version-controlled scripts. An anonymized reviewer reproduction package can be supplied during peer review. Two scripts carry this paper's experiment: one corpus pass writes the retrieval layer, and one analysis runs every arm. No judgment text is stored anywhere: the retrieval layer holds unordered bags of hashed tokens, which is

what BM25 consumes and which cannot be read back as a passage. Three guarantees are enforced in code rather than asserted:

- **Determinism.** Every analysis is verified byte-identical across runs; all output orderings break ties on a stable key.
- **Figure tracing.** Every figure quoted in this manuscript is declared in a guard script with the results file and key it comes from, and matched as an exact string. The guard fails if any figure drifts.
- **Freshness.** A stamp file hashes the code each result depends on transitively, so a result that predates the code that produced it is refused.

Baselines used in this paper are frozen with the repository head at which they were computed, and the freeze scripts refuse to overwrite. Every random arm — the 20 matched-random draws per fold, the 10 volume-control draws per freeze horizon, and the query sample itself — is drawn from a seed recorded in the results file, and the whole experiment was re-run from scratch and verified to produce byte-identical output.

Generative-AI systems were used as research-assistance tools for code drafting, literature-search assistance, manuscript drafting/editing, and adversarial critique under the author's supervision. They were not used as authors or as independent peer reviewers, and no model-generated prose or judgment defines a primary endpoint. Quantitative claims in the manuscript are traced to the version-controlled result artifacts described above; the author retains responsibility for the research question, claim boundaries, interpretation, and final submission.

12. Conclusion

We tested whether a recurrence-based corpus-cleaning intervention changes a legal retrieval system beyond what would follow from removing the same amount of evidence at random. On 105575 resolved statutory citations, it does in this corpus: targeted removal costs 2.7 times the mean same-volume loss, and in the five folds with the largest removals it is worse than all 20 matched draws. The removed contexts are therefore disproportionately useful to this retrieval instrument despite their recurrence.

On a different downstream task in the same corpus, the same intervention is volume-equivalent. The paired result is the main evaluation lesson: the effect of preprocessing depends on both the intervention and the task, and a matched-volume arm can distinguish a class-specific effect from a quantity effect.

Applied to staleness, the same comparison leaves 70%, 64%, and 62% of the one-, two-, and four-quarter MRR loss associated with age after controlling for index shrinkage, subject to the publication-composition limitation in §8.

Matched-budget and temporal controls are established prior art. The paper's contribution is the controlled legal-retrieval evidence above and the age-versus-shrinkage estimand on this corpus. We recommend reporting a same-volume control when legal-AI preprocessing removes substantial evidentiary volume before attributing a downstream change to the class of text removed.

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